

THE DYNAMICAL EVOLUTION OF THE SHORT-PERIOD EXTRASOLAR PLANET AROUND ν ANDROMEDAE IN THE PRE-MAIN-SEQUENCE STAGE

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ABSTRACT

We study the dynamical evolution of short-period planets in the multiple planetary system during the epoch of protostellar disk depletion. Through a detailed case analysis for the triple-planet system around ν Andromedae, we identify the necessary condition for the survival of its short-period planet b. In this study, we calculate the planets' orbit evolution including effects of the post-Newtonian potential of the host star, the potential of an evolving disk, and the potential due to the flattening of the star produced by the stellar rotation and the tide from the planet. At its present-day semimajor axis of 0.059 AU, the negligible eccentricity of planet b is most likely damped by the dissipation of the tidal perturbation induced by the star on the planet. But this process also leads to the heating of the planets' interior and probably the inflation of their sizes. If planet b once had an eccentricity greater than 0.1 at its present location, the tidal dissipation process would induce it to overflow its Roche lobe, lose its mass, and undergo an orbital expansion. Using this constraint, we reconstruct the eccentricity evolution of planet b. We find that even if planet b arrives at its present location on a nearly circular orbit, its eccentricity could have been excited by the sweeping secular resonances of two outer planets, c and d, during the disk depletion. In addition to the contribution from the disk potential, the resonant condition is also modified by the precession of planet b due to the post-Newtonian correction and rotational distortion of ν And's gravitational potential. Today, the former effect offsets the secular resonance, whereas the latter effect is weak. But during the depletion of the disk, precession due to the relativistic correction is outpaced by that due to the disk potential. The passage of the sweeping secular resonance near the surface of ν And cannot be avoided unless it had a sufficiently fast spin to provide a flattened shape with a finite quadrupole moment, which dominates the precession of the planet's orbit. Finally, we show that the survival of planet b requires its eccentricity to be low at all times, which would be possible only if the spin period of ν And was shorter than 2 days during the depletion of the disk.

Subject headings: celestial mechanics — planetary systems: formation —
 planetary systems: protoplanetary disks — stars: individual (ν Andromedae)

1. INTRODUCTION

Radial velocity surveys of approximately 10^3 stars have led to the discovery of over 100 planets.³ Due to observational selection effects, most of these planets have masses comparable to that of Jupiter. They also have a nearly logarithmic period distribution ranging from a few days to several years. The current theory of sequential planet formation suggests that gas giants are preferentially formed near the snow line (Ida & Lin 2004b) and that some close-in planets have migrated to their present-day orbits through planet-disk tidal interaction (Lin & Papaloizou 1986). While migration of some planets halts with periods of a few days, probably due to their tidal or magnetic interaction with their host stars (Lin et al. 1996), those with periods of weeks to months may have stalled as a consequence of disk gas depletion (Trilling et al. 1998; Armitage et al. 2002). However, since the disk depletion timescale is generally much longer than the planets' migration timescale in the innermost disk region, only a small fraction of the migrating planets can attain week- to month-long periods. Nevertheless, the observed ratio of frequencies between these two populations can be accounted for if we assume that a large population of planets that migrated to near their host stars may have perished (Ida & Lin 2004a).

In order to migrate from their birthplace near the snow line to near their host stars, short-period planets must emerge during the phase of active disk evolution. In such disks, gap formation induced by the tidal torque of the emerging protoplanet leads to the accumulation of grains and planetesimals and promotes the formation of additional planets beyond its orbit (Bryden et al. 2000). Most known planet-bearing stars, especially those with close-in planets, show signs of additional planets. There are at least 14 stars with well-determined multiple-planet kinematic structures. Among these multiple-planet systems, that around ν Andromedae (Butler et al. 1999) is particularly interesting because it contains one short-period and two intermediate-period planets. The longitudes of the periastra of the outer two planets are almost aligned, which makes the system more stable in the long run (e.g., Laughlin & Adams 1999; Rivera & Lissauer 2000; Stepinski et al. 2000; Barnes & Quinn 2001; Ito & Miyama 2001; Lissauer & Rivera 2001; Chiang et al. 2001).

In this paper, we present a case study of the early evolution of the planetary system around ν And. We determine the necessary conditions that preserve the innermost planet. Based on this analysis, we demonstrate that the survival of the short-period planets in multiple-planet systems requires rapid stellar spins and that around most host stars, they are vulnerable to tidal disruptions during the depletion of protostellar disks.

Similar to other extrasolar planets with orbital periods of less than a week, ν And b has a negligible eccentricity. The orbits of these “hot Jupiters” are most likely to be circularized by the dissipation of the stellar tidal perturbation within their envelopes (Rasio et al. 1996). In addition, their orbits are affected by

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³ See <http://exoplanets.org/almmacframe.html>.

the relativistic correction to the Newtonian gravitational potential of their host stars. Other distortions induced by the rotational or tidal flattening of the host stars also lead to precession. The orbital period of ν And b is 4.617 days, and the upper limit of its eccentricity is ~ 0.01 . If it has any residual eccentricity today, the longitude of ν And b's periastron would advance, analogous to Mercury in the solar system, mostly due to the relativistic correction of the Newtonian potential (Mardling & Lin 2002).

The effect of the stellar flattening due to its rotation and tidal interaction with the planet also has an effect on the motion of the short-period planets (Mardling & Lin 2005). With a current spin period of $S_* = 10.2$ days, the rotation of ν And has little effect on the motion of planet b (see § 2). However, the observed rotation periods of pre-main-sequence stars have a uniform distribution starting with 0.5 days (Stassun et al. 1999). Since the precession rate of the periastron due to the J_2 component of the rotating host star is proportional to S_*^{-2} (see § 2), the effect of the rotation may have had a much stronger influence in the early evolutionary phase of ν And when it rotated much faster.

Nearly all pre-main-sequence stars are surrounded by protostellar disks (Haisch et al. 2001). The potential of these disks also causes the advancement of the periastron (Ward 1981; Nagasawa et al. 2003). In addition, the secular interaction between members of multiple-planet systems also leads to precession of the periastron, which self-consistently modulates their exchange of angular momentum and eccentricity variations.

Within the limit of the classic celestial four-body interaction, the orbit of ν And b is nearly in a secular resonance with its sibling planet ν And d. The orbital integration of the three planets and host star shows that around a point-mass stellar potential, the planets' secular interaction would induce the eccentricity of ν And b to vary with an amplitude of 0.15 over a period of $\sim 10^5$ yr. But beyond the scope of classical mechanics, the post-Newtonian correction to the stellar potential shifts the condition of the secular resonance and suppresses the large-amplitude oscillation of the eccentricity of ν And b (Mardling & Lin 2005). However, during its pre-main-sequence phase, the contribution of a protostellar disk around ν And also changed the location of the secular resonances. The amount of shift is determined by the surface density of the disk gas. During its depletion, the location of the planetary system's secular resonances sweep throughout the disk, and the eccentricity of the planets along the paths can be excited (Ward et al. 1976; Heppenheimer 1980; Nagasawa et al. 2000, 2001, 2003). Here we show that if ν And had a slow initial spin frequency, the secular resonance of ν And d would sweep past the semimajor axis of ν And b and would excite its eccentricity to above 0.2. Although the orbit of ν And b may be circularized within the subsequent megayears, the dissipation of the stellar tidal disturbance would lead to its intense internal heating, radial expansion, Roche-lobe overflow, and mass loss (Gu et al. 2003). Upsilon And b would either be tidally disrupted or migrate outward and would not be able to attain its present-day short period.

In this paper, we find the necessary condition for the survival of hot Jupiters by analyzing their orbital evolution in multiple planetary systems. We focus primarily on the ν And system and determine the eccentricity evolution of planet b, including the effects of the depletion of the protostellar disk, post-Newtonian gravity, and the stellar rotation. In § 2, we calculate the current orbits of planets in the ν And system. In § 3, we show the formulae of force caused by the other planets, the gravity of the protostellar disk, the post-Newtonian potential, and the stellar rotation that we used in our numerical simulations. To help the

TABLE 1
ORBITAL PARAMETERS OF ν AND PLANETS

Planet	a (AU)	e	ϖ (rad)	$M \sin i$ (M_J)	T_{peri} (JD 2,450,000+)
b.....	0.059	0.012	1.3	0.69	2.093
c.....	0.829	0.28	4.36	1.89	160.5
d.....	2.53	0.27	4.54	3.75	64

NOTE.—See <http://exoplanets.org/almamacframe.html>.

understanding of our numerical simulations, we also present the advancing rates of the planets' periastra and the eigenfrequencies of the system. In § 4, we study with numerical simulations the evolution of planet b during the disk depletion. In § 5, we summarize our results and propose that the presence of planet b is possible only if the rotation period of ν And was shorter than 2–3 days in its early evolutionary stage.

2. CURRENT ORBITAL PROPERTIES OF THE ν AND SYSTEM

The observationally inferred orbital elements of the ν Andromeda planets are shown in Table 1. The mass of the host star is $M_* = 1.3 M_\odot$, where $M_\odot$ is the solar mass. In this paper, a_j is the semimajor axis (AU), e_j is the eccentricity, ϖ_j is the longitude of periastron (radian), and T_{0j} is the time of periastron passage (from Julian Date [JD] 2,450,000), M_j is the mass of planet j ($j = b, c$, and d , for planets b, c, and d, respectively), M_J is the Jovian mass, and i is the inclination angle of the orbital plane to the line of sight from the Earth. Throughout this paper, we assume that the planets around ν And are orbiting in the same plane. We also assume that these systems are viewed edge-on and that their masses correspond to their minimum values ($\sin i = 1$). Our analysis is essentially independent of the angle of orbital plane. But a nonnegligible variation in the spin vector of their orbit could introduce a significant modification to the analysis presented here.

Starting from these values, we calculate evolution of the eccentricities of the three planets by direct orbital integrations (Fig. 1a). In this numerical simulation, only the mutual gravity between planets and the host star is included. Although ν And has a binary companion M4 star at 750 AU (Lowrance et al. 2002), the effect of this binary companion on the planets is negligible. We neglect the gravity from the binary companion in this paper.

In Figure 1a, we show the numerical solutions that were obtained with a point-mass potential for the host star. The numerical solutions show that planets c and d are locked in librating orbits. (Libration means that the difference between periastra is confined within a certain range less than π , and circulation means that the relative longitude of periastra modulates over the full range of phases from $-\pi$ to π .) These solutions indicate that the relative longitude of periastron between c and d is $|\varpi_c - \varpi_d| \lesssim 1$ rad. They also indicate that the periastron of planet b is librating around a phase that shifted 180° from the periastra of the other planets.

In the absence of the relativistic correction to the potential of the host star, ν And b is near the secular resonance of its two planetary siblings (see § 3.5). In this case, the eccentricity of planet b can grow to 0.2. With such a large eccentricity, the planet's internal tidal dissipation rate is high, and the planet is vulnerable to tidal disruption (Gu et al. 2003). In reality, however, the effect of the post-Newtonian force from the host star (see § 3.3) changes the precession rate of ϖ , and the enhancement

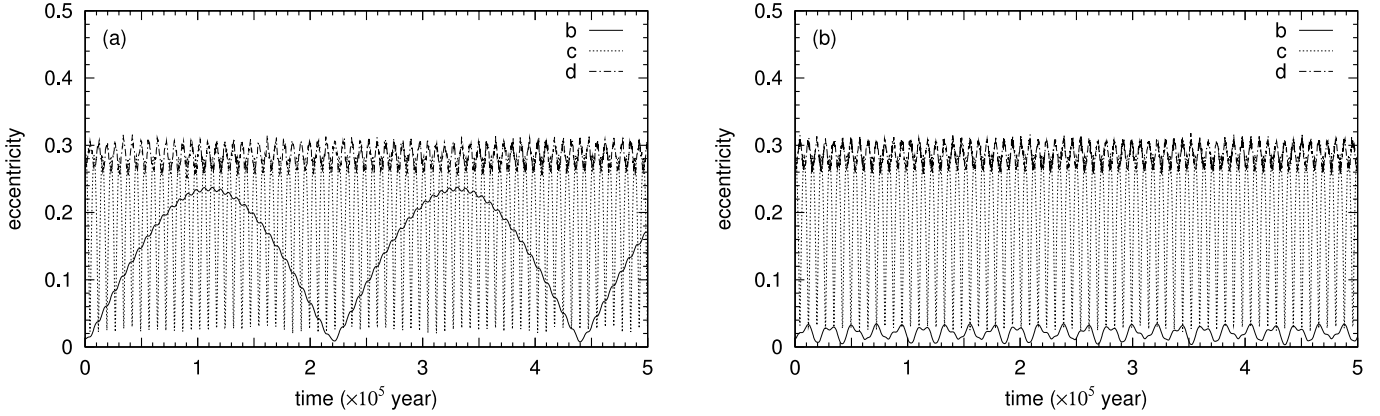


FIG. 1.—(a) Eccentricity evolution of three planets around ν And with a point-mass potential. The solid, dotted, and dash-dotted lines show the evolution of planets b, c, and d, respectively. (b) Planetary eccentricity evolution around ν And including the post-Newtonian effect.

of the eccentricity is suppressed (Fig. 1b). Today, the small amplitude of eccentricity variation during the libration does not lead to adequate tidal dissipation within planet b to significantly affect its internal structure.

3. FORCE AND PRECESSION RATES

In this section, we consider the precession due to the disk potential. Although this contribution does not exist today, it dominates the precession of all the planets during the early dynamical evolution of this planetary system.

All planets of ν And are subject to the gravitational force from their sibling planets, the host star, and the protostellar disk. Although the contribution of the post-Newtonian force and the force due to the rotational distortion of its host star is negligibly small for the outer planets, they are important for the innermost planet b. We briefly recapitulate the nature of these forces and their effects on the orbital evolution of the planets. We derive an analytic solution for the precession rates of the planets. Following the standard first-order perturbation theory, the precession rate is averaged over the orbit and expanded to the lowest order of the eccentricity (e.g., Brouwer & Clemence 1961; Murray & Dermott 1999).

3.1. Three-Planet Interaction

The gravitational force on the planet i from the other planet j is

$$\mathbf{f}_{i,\text{planet}} = G \sum_{j \neq i} \left[M_j \left(\frac{\mathbf{r}_j - \mathbf{r}_i}{|\mathbf{r}_j - \mathbf{r}_i|^3} - \frac{\mathbf{r}_j}{|\mathbf{r}_j|^3} \right) \right], \quad (1)$$

where G is the gravitational constant and $\mathbf{r}$ is the position vector from the star. To the lowest order of the eccentricity, the precession rate of the periastron is

$$\frac{d\varpi_{i,\text{planet}}}{dt} = \sum_{j \neq i} \frac{1}{t_{i,j}} \left[1 + C_{i,j} \left(\frac{e_j}{e_i} \right) \cos(\varpi_i - \varpi_j) \right], \quad (2)$$

where

$$\frac{1}{t_{i,j}} \equiv \frac{GM_j \alpha b_{3/2}^{(1)}}{4n_i a_i^2 \max(a_i, a_j)}, \quad (3)$$

$C_{i,j} \equiv -b_{3/2}^{(2)}/b_{3/2}^{(1)}$, $b_i^{(k)}$ are the Laplace coefficients whose variables are $\alpha = \min(a_i, a_j)/\max(a_i, a_j)$, and n_i is the mean motion of planet i . In equation (A3), we expand the relative distance between two interacting planets. We show in the Appendix that the coefficient of the low-order term in eccentricity $F_1 = \alpha b_{3/2}^{(1)}/4$ and $F_3 = -\alpha b_{3/2}^{(2)}/4$.

With these equations we now consider the orbits of ν And b, c, and d. From equation (2), we find the time-averaged precession rate of the periastron to be $\sum_{i \neq j} (1/t_{i,j})$ in circulating orbits with respect to all other planets. For planet b, $d\varpi_{b,\text{planet}}/dt$ is estimated to be $43'' \text{ yr}^{-1}$.

Without the cosine term in equation (2), the total precession rates (including the secular contribution from planet b) for planets c and d are estimated to be $d\varpi_{c,\text{planet}}/dt = 180'' \text{ yr}^{-1}$ and $d\varpi_{d,\text{planet}}/dt \sim 51'' \text{ yr}^{-1}$. When it is included, the cosine term periodically dominates the rates and reverses the direction of planet c's precession. The magnitude of both e_c and e_d also modulates. From equation (2) we find that the precession rate induced by ν And c and d on each other is identical and that $d\varpi_c/dt = d\varpi_d/dt = 51'' \text{ yr}^{-1}$ when $e_c/e_d \cos(\varpi_c - \varpi_d) = 0.54$. In the numerical solutions presented in § 2, planets c and d librate around each other's orbit with the same mean precession rate of $\langle d\varpi_c/dt \rangle \sim \langle d\varpi_d/dt \rangle \sim 50'' \text{ yr}^{-1}$ and with $\langle e_c/e_d \cos(\varpi_c - \varpi_d) \rangle \sim 0.57$.

Our numerical results indicate that the mean precession rates of all three planets are relatively close, but the dominant effects that regulate the eccentricity modulation are quite different. In a system with N planets, there are N eigenfrequencies for the coupled oscillators (see § 3.5). In equation (2), the value of $e \sin \varpi$ changes with time in the linearly combined form of these N eigenfrequencies. The averaged precession rates of the periastra of planets are almost the same as one of the eigenfrequencies of the system that is derived from the secular perturbation theory. If the values of two eigenfrequencies are nearly equal to each other, the precession rates of the corresponding planets become very similar, and their amplitude of eccentricity variation also becomes large. When the coefficient associated with one particular eigenfrequency becomes dominant, the apsidal motion is determined by that frequency, and the precession rates also attain similar values.

But in this case, the orbits of the planets undergo libration, and there is no strong secular perturbation between them (see also the Appendix). Since the eigenfrequency of planet b ($43'' \text{ yr}^{-1}$) is close to one of the others ($46'' \text{ yr}^{-1}$), its eccentricity becomes large, as seen in Figure 1a. On the other hand, planets c and d are

librating relative to each other, and their eigenfrequencies are not close to each other (46 and $210'' \text{ yr}^{-1}$). One of the eigenfrequencies is close to that of planet b, but the secular perturbation induced on planets c and d by b is small. Therefore, the mean values of e_c and e_d do not significantly increase.

3.2. Effect of the Gravity of the Protostellar Disk

In the presence of a protostellar disk, a planet's motion is greatly modified by its gravitational potential. Nagasawa et al. (2003) showed that the disk depletion has induced a strong secular perturbation between planets c and d that led to a net exchange of their angular momenta and modification of their eccentricities.

Here we adopt the same model for the disk potential that was introduced by Nagasawa et al. (2003). A thin and axisymmetric disk is assumed to be in the orbital plane of the planets. We only consider its contribution to the axisymmetric component of the gravitational potential. We neglect the nonaxisymmetric torque and do not consider the disk's response to the planets' perturbation. **We assume that the disk inside the orbit of planet d has been severely depleted by the stage when the three planets acquired most of their masses.** This conjecture is based on the numerical simulations that showed that the protoplanets can tidally induce the formation of a gap near their orbit after their growth through gas accretion (Goldreich & Tremaine 1980; Lin & Papaloizou 1980, 1986, 1993; Takeuchi et al. 1996). The surface density (Σ) of the disk outside of planet d is set to be proportional to $r^{-3/2}$, where r is the planar distance from the host star.

In the calculations below, we show that the dominant effect of the sweeping secular resonances occur when the mass of disk just beyond the orbit of planet d becomes comparable to that of its mass. At this late stage, our disk model provides an adequate approximation for the matter distribution. Although the disk's dynamical response to the planets' perturbation and the disk's torque can lead to nonnegligible eccentricity evolution for planet d, it is too far removed from v and b to be directly effective. Our approximation is adequate for illustrating the effect of sweeping secular resonance on planet b (see also § 4.6.2).

The force from the disk whose inner edge is located at d ($d > a_d$) (Heppenheimer 1980; Ward 1981) is given by

$$f_{i,\text{disk}} = 8\pi G \Sigma(r_i) \frac{r_i}{r_i} \sum_{n=0} \left[\frac{B_n}{2n+1} \left(\frac{r_i}{d} \right)^{2n+1/2} \right], \quad (4)$$

where the coefficient $B_n \equiv [(2n)!/2^{2n}(n!)^2]^2 n(2n+1)/(4n+1)$. In the numerical simulation, we take the initial surface density of the disk to be 5 times that of the minimum mass solar nebula model (Hayashi 1981) and put $d = 4.5 \text{ AU}$. Our results are not sensitive to the initial conditions, since the sweeping secular resonances occur when the disk mass just beyond d declines below some particular value.

The averaged precession rate due to this force is

$$\frac{d\varpi_{i,\text{disk}}}{dt} = \frac{4\pi G \Sigma(a_i)}{n_i a_i} \sum_{n=1} \left[B_n \left(\frac{a_i}{d} \right)^{2n+1/2} \right]. \quad (5)$$

The effect of the disk's potential is stronger for the outer planets. The contribution of the disk to the precession rates of planets b, c, and d are estimated as $0''.64$, $35''$, and $265'' \text{ yr}^{-1}$, respectively.

The precession rate of periastra under the perturbation of the disk are given by the summation of equations (2) and (5).

The forces and the precession rates of periastra due to the disk decrease with the depletion of the disk. In § 4, we consider the case that the disk is depleted exponentially on the timescale τ such that

$$\Sigma(t) = \Sigma(t=0) \exp(-t/\tau). \quad (6)$$

The actual evolution pattern of the disk depletion and the surface density profile of the disk are not so important in the present context, since we are primarily interested in the evolution of planet b due to the sweeping secular resonances. The disk's potential only contributes to planet b indirectly through the motion of planets c and d. Nagasawa et al. (2003) calculated the orbital evolution of planets c and d with several different disk profiles and two prescriptions for depletion and found that there is no fundamental difference between the results obtained with different models of the disk.

3.3. Effect of the Post-Newtonian Correction on the Gravity of the Host Star

The force induced by the post-Newtonian gravity from the host star given in Kidder (1995) is

$$\begin{aligned} f_{i,\text{rela}} = & -\frac{G(M_* + M_i)}{r_i^2 c^2} \\ & \times \left\{ \left[(1 + 3\mu)v_i^2 - 2(2 + \mu) \frac{G(M_* + M_i)}{r_i} - \frac{3}{2} \mu \dot{r}_i^2 \right] \frac{\mathbf{r}_i}{r_i} \right. \\ & \left. - 2(2 - \mu) \dot{r}_i \mathbf{v}_i \right\}, \end{aligned} \quad (7)$$

where c is the light speed, $\mathbf{v}$ is the velocity vector, $v = |\mathbf{v}|$, $\dot{r}_i = dr/dt$, and $\mu \equiv M_* M_i / (M_* + M_i)^2$.

To the lowest order of the eccentricity, the precession rate of the periastron is expressed as (Mardling & Lin 2002)

$$\frac{d\varpi_{i,\text{rela}}}{dt} = \frac{3n_i G M_*}{a_i c^2}. \quad (8)$$

The post-Newtonian effect is important only to the short-period planets. Although the advance rate of the periastron of planet b due to the relativistic correction is $67'' \text{ yr}^{-1}$, those of planets c and d are only $0''.09$ and $0''.006 \text{ yr}^{-1}$, respectively.

3.4. Effect of the Rotation of the Host Star

The gravity of the host star deviates from a spherical symmetry due to its rotational distortion. The J_2 component introduced by this rotation induces an advancement in the periastra of accompanying planets. Assuming that the axis of the rotation is perpendicular to the orbital plane of the planets, the force on the planets is (Eggleton & Kiseleva 1998)

$$f_{i,\text{spin}} = -\frac{R_*^5 (1 + M_i/M_*) k_* \Omega_*^2}{2r_i^4} \frac{\mathbf{r}_i}{r_i}, \quad (9)$$

where R_* is the radius of the host star, Ω_* is the spin rate of the host star, and $k_* = 0.02$ is the tidal Love number for a star with polytrope of index 3 (Goldreich & Soter 1966).

The precession speed of the periastron caused by this force is

$$\begin{aligned} \frac{d\varpi_{i,\text{spin}}}{dt} &= \frac{k_* n_i}{2} \left(1 + \frac{M_i}{M_*}\right) \left(\frac{R_*}{a_i}\right)^5 \left(\frac{\Omega_*}{n_i}\right)^2 \\ &\sim 135 \left(\frac{a_i}{0.05 \text{ AU}}\right)^{-7/2} \left(\frac{M_*}{M_\odot}\right)^{-1/2} \\ &\quad \times \left(\frac{k_*}{0.02}\right) \left(\frac{R_*}{R_\odot}\right)^5 \left(\frac{S_*}{1 \text{ day}}\right)^{-2} \text{ arcsec yr}^{-1}, \quad (10) \end{aligned}$$

where $S_* = 2\pi/\Omega_*$ is the rotation period of the host star. The precession speed strongly depends on the distance from the host star. The precession rates of planets c and d induced by the rotational distortion of the host star are $<10^{-4}$ times that of planet b.

The actual magnitude of planet b's periastron-precession rate is $d\varpi_{b,\text{spin}}/dt \sim 270(\text{day}/S_*)^2 \text{ arcsec yr}^{-1}$ for $R_* = 1.26 R_\odot$. The current rotation period of ν And is $S_* = 10.2$ days. Using this value we obtain $d\varpi_{b,\text{spin}}/dt \sim 3'' \text{ yr}^{-1}$, which is negligible compared with the precession induced by the perturbation due to other planets. However, solar-type stars are observed to spin down during the early phases of their main-sequence evolution (Soderblom et al. 1993). It is entirely possible that ν And may once have had a rapid spin that significantly modified its precession frequency. For example, the precession due to a 1.5 day spin period is $d\varpi_{b,\text{spin}}/dt \sim 120'' \text{ yr}^{-1}$, which exceeds even that due to the post-Newtonian gravity. Equation (12) indicates that $d\varpi_{i,\text{spin}}/dt$ is a sensitive function of R_* . The radius of ν And ($M_* = 1.3 M_\odot$) in the pre-main-sequence stage evolves from ~ 4 to $\sim 1.3 R_\odot$ in about $3 \times 10^7 \text{ yr}$. The stellar contraction is rapid at the beginning of evolution, and after about 3 Myr, the radius reduces slowly from ~ 1.5 to $\sim 1.2 R_\odot$. We consider the effects of radius change and the spin-up related to it in § 4.5. Unless stated otherwise, we assume a constant stellar radius with the main sequence for the mass of ν And, i.e., $R_* = 1.26 R_\odot$, in this paper.

Finally, we also consider the tidal distortion induced by the presence of planet b. This effect is important only to the precession of planet b. This force can be expressed as (Goldreich & Soter 1966)

$$f_{b,\text{tide}} = -\frac{12GM_b R_*^5 (1 + M_b/M_*) k_*}{r_b^8} r_b. \quad (11)$$

When the rotation speed of the host star is less than 60 days, $f_{b,\text{tide}} < f_{b,\text{spin}}$. Therefore, the precession due to this force is almost negligible in the ν And system.

3.5. Analytical Approaches

The total precession rates of periastra are represented by the linear summation of these four contributions in equations (2), (5), (8), and (12). Among these four rates, only the contribution from the disk $d\varpi_{i,\text{disk}}/dt$ is time-dependent. Since the ratio between $d\varpi_{i,\text{disk}}/dt$ and the total precession rate is different in each planet, the latter declines at a different pace for each planet. Consequently, during a stage of the disk depletion, there is a possibility that the precession rates of some planets match each other. As they pass through these resonances, the participating planets induce strong secular perturbations on each other, and consequently large changes in eccentricities occur.

Before illustrating the results of our numerical simulations, we present some analytical estimates using a modified version

of the secular perturbation theory that enables us to treat the evolution of planets with modest eccentricities. Since the outer two planets are on librating orbits, the analysis with independent precession rates can be misleading. Instead, we consider the normal mode eigenfrequencies g of the entire system as three coupled oscillators. The magnitude of g is obtained from the equation of motion using a determinant,

$$\begin{vmatrix} A'_{i,i} - g & A_{i,j} & A_{i,\ell} \\ A_{j,i} & A'_{j,j} - g & A_{j,\ell} \\ A_{\ell,i} & A_{\ell,j} & A'_{\ell,\ell} - g \end{vmatrix} = 0. \quad (12)$$

The coefficients $A'_{r,r}$ and $A_{r,s}$ are functions of the semimajor axes, mass, and eccentricities of the planet r and s (see eqs. [A21], [A19], and [A33] in the Appendix). When the precession of the planets' orbits are circulating with respect to each other, the $A_{r,r}$ values are close to the periastron's precession rates of planet r .

Due to its small mass and semimajor axis, planet b's secular perturbation on the evolution of planets c and d is negligibly weak. Thus, the normal mode of planets c and d is mostly regulated by their mutual secular interaction. The eccentricity modulation of planets c and d during the disk depletion can be derived from the evolution of the libration center in their Hamiltonian contours (Nagasawa et al. 2003). Putting $e_c/e_d = x_m$, $\gamma = \alpha^{1/2}(M_c/M_d)$, and $\alpha = a_c/a_d$, the eccentricity ratio of the libration center (x_m) is obtained from

$$x_m^2 + \frac{\Delta N + 1 - \gamma}{\gamma C_{c,d}} x_m - \frac{1}{\gamma} = 0, \quad (13)$$

where

$$\Delta N = \frac{8\pi\Sigma(a_c)a_d^2}{M_d b_{3/2}^{(1)}} \sum_{n=0} B_n \left(\frac{a_c}{d}\right)^{2n+1/2} (1 - \alpha^{1/2-2n}). \quad (14)$$

Without the disk ($\Delta N = 0$), the ratio of the eccentricities at libration center is $x_{m0} = 0.52$. Considering the ratio of averaged eccentricities of planets c and d is the same as the eccentricity ratio of the libration center (x_m), the eccentricity of planet c is written using e_d as $e_c = x_m e_d$. From the conservation of total angular momentum, $e_d^2 = (1 + x_{m0}^2 \gamma)/(1 + x_m^2 \gamma) e_{d0}^2$, where e_{d0} is the current eccentricity of planet d. Using $e_{d0} = 0.28$, we plot e_c and e_d in Figure 2a with different disk densities. The unit of the x -axis is the density of the minimum mass nebula (ρ_H).

The main constraint of our calculation is the present values of e_c and e_d . If we start the computation with an initial surface density equal to 5 times that of the minimum mass disk, the value of $e_c = 0.52$ and $e_d = 0.08$ with $x_m = 6.48$. (The value of e_c would be much smaller with a less massive initial disk mass.) Since this value of x_m is at the libration center, we take into account the libration width and use $e_c \sim 0.57$ and $e_d \sim 0.04$ for the initial condition of the numerical simulations to be presented in § 4.

In Figure 2b, we plot the precession speeds of periastra that correspond to the eccentricities calculated from equations (A26), (A27), and (A29). In this figure, we assume that planets c and d were initially on librating orbits with respect to each other and that planet b's orbit was circulating with respect to those of the other two planets. If planet b is in a librating state today, its precession rate would approach that of the outer planets. The magnitude of the precession rate is sensitively determined by

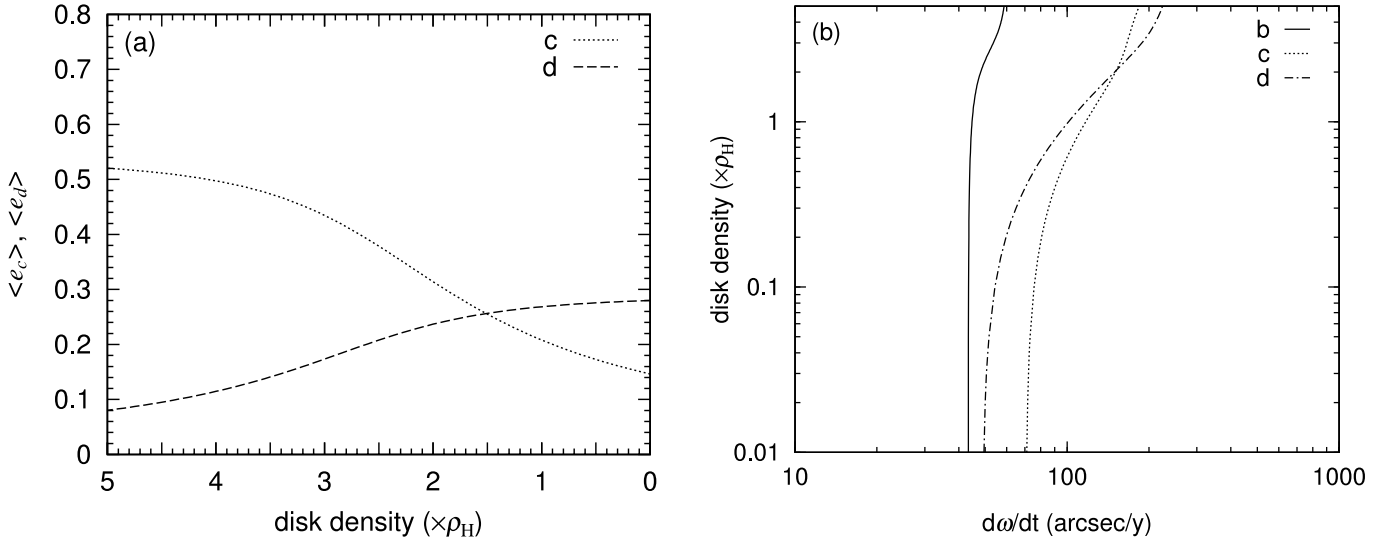


FIG. 2.—Evolution of the planets as a function of the disk density obtained from analytical estimation. The solid, dashed, and dash-dotted lines show the evolution of planets b, c, and d, respectively. The units of the disk density are normalized to the density of the minimum mass nebula. The effects of stellar rotation and the post-Newtonian potential are not included. Panel (a) illustrates the eccentricities of planets c and d. Panel (b) shows the apsidal motion of three planets corresponding to the eccentricities of (a). We assume here that planets c and d are librating with $\varpi_c - \varpi_d = 0.64$, and planet b is orbiting with eccentricity $e_b = 0.01$.

the eccentricity of outer planets and their libration amplitude. The necessary condition for planet b to enter into libration requires a suitable range of other planets' eccentricities during each stage of the disk depletion.

In Figure 3, we show the eigenfrequencies obtained from equation (14) with the above inferred initial eccentricities. The average value $\langle \cos(\varpi_c - \varpi_d) \rangle = 0.8$ is used in these calculations. The thick dash-dotted line represents the case in which we included only the contribution from the gravity of three planets, the disk, and a point-mass stellar potential. The dashed line represents the case in which the post-Newtonian effect is added to the stellar potential. The approximately perpendicular solid line represents the case in which we also included the effect of rotational flattening with the current stellar spin period ($S_* = 10.2$ days). The dotted, dash-dotted, and double-dashed lines represent the cases of $S_* = 3, 2$, and 1 day, respectively. The horizontal dashed line indicates the initial disk surface density, which is 5 times that of the minimum mass disk.

These solutions of equation (14) yield three sets of eigenvalues, g_1 , g_2 , and g_3 . The eigenvalues are not exactly the precession frequency of planets, but eigenfrequencies g_1 and g_2 are associated with the orbital evolution of planets c and d. The precession and eccentricity modulations of planets c and d are not significantly affected by the variations in the post-Newtonian potential and stellar rotation. In all our models, g_1 and g_2 follow nearly identical evolutionary tracks, and their near-degeneracy makes these overlapping lines appear as thick solid lines. The remaining eigenvalue g_3 is closely associated with the average precession frequency of planet b. Since planet b is dynamically separated from the other planets and has a relatively low mass, the magnitude of g_3 of the system is almost the same as the averaged precession rate of the planet b, when its precession is circulating relative to that of the other planets. (Planet b's precession frequency has a significant difference from the eigenfrequency of the system g_3 if it has a librating orbit relative to the other planets.)

Near ν And, the precession rate of planet b is sensitively affected by both the post-Newtonian contribution and the rotational distortion of the star. In order to highlight these differences,

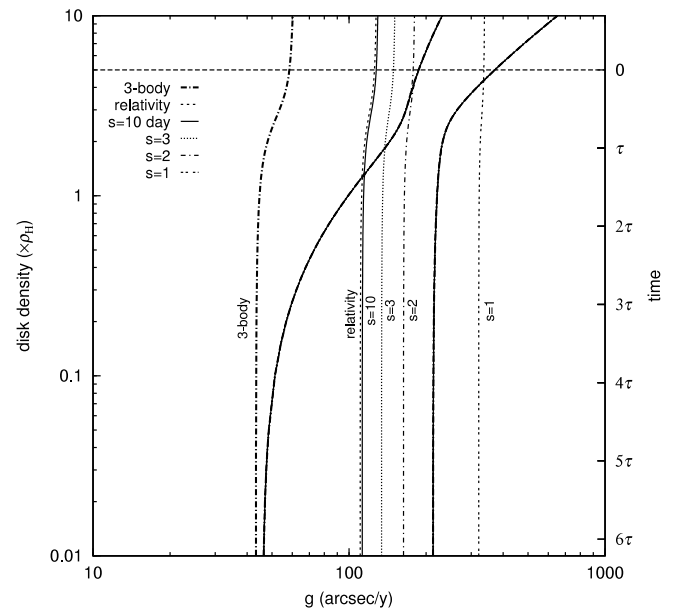


FIG. 3.—Changes of the eigenfrequencies of the system during the disk depletion. The approximately perpendicular lines represent g_3 values in various models. The two slanting lines represent the g_1 and g_2 values due to the secular perturbation of planets c and d. Since g_1 and g_2 are not significantly affected by the stellar potential, they have very similar values for each model. These overlapping values contribute to the thick solid lines. The thick dash-dotted line represents g_3 due to the three-planet interaction in an evolving disk potential and a point-mass potential for ν And. The dashed line is for the case in which the post-Newtonian effect is included in the stellar potential. The horizontal dashed line indicates the epoch when the disk density is everywhere 5 times that of the minimum mass disk. The remaining lines represent the cases that take into account effects of both the post-Newtonian and stellar rotation. The solid, dotted, dash-dotted, and double-dashed lines correspond to the cases of $S_* = 10.2, 3, 2$, and 1 day, respectively. The vertical axis on the right shows the corresponding time under the assumption that the exponential decay of the disk started from 5 times that of the minimum mass disk.

a range of g_3 values is obtained for different models. For example, today, with only the three-body interaction (i.e., no disk), the asymptotic value of g_3 would be close to that of g_1 , and planet b would be in a secular resonance with the outer planets if the host star has a point-mass potential. In this case, e_b would excited to modest values (Fig. 1a). But when the effect of post-Newtonian gravity is taken into account, planet b is no longer in resonance with the outer planets (Fig. 1b).

Comparing Figures 2b and 3, we find that both the apsidal precession rates of planets c and d are close to g_1 when they are in a librating state. Therefore, strong secular perturbation comes from only g_1 . The values of the eigenfrequencies of g_1 and g_2 do not change significantly whether the planets' orbits undergo libration or circulation (see the Appendix). However, when these planets attain a state of circulation, g_2 also contributes to the oscillation of the system because the precession rate of planet c is close to g_2 . During the initial depletion of the disk, the lines of eigenfrequencies g_1 and g_2 in Figure 3 approach each other. However, eigenfrequencies of systems of two planets with finite masses cannot become exactly degenerate. (In Fig. 3, g_1 approaches but never crosses the value of g_2 .) In this strict sense, there is no secular resonance between them. Nevertheless, the precession rates of these planets (which are different from g_1 and g_2) can episodically be equal to each other.

During the depletion of the disk, g_3 decreases much less than g_1 and g_2 because we have adopted the Σ -distribution of a minimum mass nebula. Since the surface density is truncated interior to 4.5 AU, the contribution of the extended disk regions to the overall gravitational potential is relatively weak for the close-in planets. The slight decline of g_3 is primarily caused by the changes in e_c and e_d rather than that of the disk mass. The amount of g_3 adjustment due to this effect is much smaller than the differences of g_3 between various models, i.e., the contribution due to relativity and stellar spin.

In models with $g_3 < g_1$ initially, the rapid decline of g_1 reduces their magnitude differences during the early stages of disk depletion. For models with the stellar spin period greater than 2 days, the values of g_1 and g_3 cross each other during the epoch $0.3 < \tau < 1.3$. When two or more eigenfrequencies of the system approach each other, the adiabatic change in the gravitational potential leads to the large modification in planet eccentricities. The magnitude difference between g_i and g_j becomes minimum when $A_{i,i} = A_{j,j}$. For practical purpose, we refer to such a configuration as an effective secular resonance. For the ν And system, the passage of this secular resonance occurs regardless of any changes in e_d due to planet d's tidal interaction with the outer disk. The magnitude of e_b changes markedly when the precession rate of planet b coincides with that of the outer planets (i.e., when $g_3 \sim g_1$).

From Figure 3, we infer that during the epoch of disk depletion, the stellar rotation needs to be faster than 2 days in order for planet b to avoid the passage of a strong resonance (in which $g_3 \sim g_1$) with the outer planets. Since the normal mode analysis in this section is only valid for small eccentricities, we perform the numerical simulations in § 4, solving full equations of motion for the present-day modest values of e_c and e_d .

4. RESULTS OF THE NUMERICAL SIMULATION

In this section, we illustrate the evolution of the eccentricities during the depletion of the protostellar disk with several sets of numerical calculations. In order to highlight the contribution from different processes, we first adopt a point-mass potential for the host star and show the evolution due to the disk depletion (set A). The effect of the post-Newtonian correction to the

gravity is analyzed separately (set B). Finally, we include other additional effects such as the host star's rotational flattening and tidal distortion induced by planet b (set C). We also show results of the case with migration of the planet d that is driven by its tidal interaction with the disk, as well as the effect of secular resonance in less complex systems with only one outer gas giant planet.

4.1. Models and Initial Conditions

We start our simulation by considering the initial surface density of the disk [$\Sigma(t = 0)$] to be 5 times that of the minimum mass solar nebula model (Hayashi 1981). Since we approximate the evolution of the disk surface density distribution by equation (6), the time after the onset of the disk depletion, t , is scaled by $\Sigma(t = 0)$ and τ . Even if the disk around ν And is initially more massive than we have assumed, the same secular interaction takes place, albeit at a different time.

We place three planets (b, c, and d) at their present-day locations, with their present-day masses, and assume that the disk is tidally truncated interior to 4.5 AU. Our numerical results do not depend critically on the assumption for the location of the disk edge.

Provided it is longer than the secular interaction timescales, the actual magnitude of the disk depletion timescale τ does not affect whether planet b evolves through secular resonance. For computational convenience, we use $(1-3) \times 10^5$ yr for the disk depletion timescale (τ). The actual disk depletion timescale is probably an order of magnitude longer than the value we have chosen for τ . To the lowest order in eccentricity, the amount of eccentricity change during the resonance passage is $\propto \tau^{1/2}$, and our numerical results indicate that the excitation of e_b is likely to be more severe than we have shown here. In § 4.5, we consider the effect of the stellar contraction on the host star's spin rate and on planet b's precession frequency. The contraction timescale we adopt in this context is also shortened in accordance with τ . With these approximations, we are able to complete all our simulations by integrating the orbits of all three planets for more than 3×10^5 yr.

In all models, we have neglected the influence of tidal interaction between the planets and the disk and assume the disk structure to be axisymmetric. The total angular momentum of the multiple-planet system is conserved, and individual planets do not migrate. This assumption is probably well justified for planets b and c, since there is little residual disk gas near their orbits. But in our models, we assumed that the disk is truncated beyond the orbit of planet d. In this case, its tidal interaction with the disk may modify the eccentricity of planet d (Artymowicz 1992; Papaloizou et al. 2001; Sari & Goldreich 2004). However, planet d's present-day semimajor axis (2.5 AU) is somewhat interior to the snow line around ν And. If gas giants preferentially formed near the snow line (Boss 1995; Ida & Lin 2004b), the modest difference between its orbital semimajor axes and the distances of the snow line would suggest that planet d probably did not migrate over extensive regions in the disk.

In order to explore the potential implications of planet d's limited migration, we adopt an ad hoc prescription. Using two sets of model parameters, we demonstrate that such a process does not significantly affect the basic outcome of our calculation. Finally, we show in § 3 the effects of the post-Newtonian gravity and stellar flattening are only effective for the planet b, and they can be neglected for planets c and d. The stellar spin period is introduced as a model parameter, and in three sets of models, we adopt it to be $S_* = 10.2, 3$, and 1 day, etc.

The orbits of planets are modified by the exchange of the angular momentum during the disk depletion (see Fig. 2a). An appropriate set of initial conditions need to be chosen so that it can lead to the current orbits of planets after the disk depletion. We use the initial values obtained from § 3.5, i.e., $e_c = 0.57$, $e_d \sim 0.04$, and $\varpi_c - \varpi_d = -0.91$. These initial conditions are not critically required. Initial values in the ranges $0.5 \lesssim e_c \lesssim 0.57$, $0 \lesssim e_d \lesssim 0.15$, and $|\varpi_c - \varpi_d| \lesssim 0.9$ can be used in combination to generate the present-day asymptotic orbital angular momentum distribution. Since the libration timescale of planets is much shorter than τ , the system adjusts adiabatically. All these combinations conserve the total angular momentum of the system and lead to the same result.

For comparison purposes, we also present results that were generated by setting the initial eccentricities to be their present-day observed values (see § 4.6.1). In all simulations, we adopt a small initial value for e_b ($=0.01$). Otherwise planet b would attain a relatively large asymptotic value of e_b except for the fortuitous case in which e_b is almost exactly eliminated during the passage of the secular resonance. After the disk depletion, the eccentric short-period planets would undergo intense tidal dissipation.

We do not include the effect of circularization due to tidal dissipation in our computation directly. Changes in the eccentricity caused by the passage of the secular resonance is faster than that due to tidal damping of the eccentricity. The timescale of the eccentricity damping due to the tidal interaction is $\propto \tau^{13/2} Q$ (Goldreich & Soter 1966). The observationally inferred Q -value for Jovian planets ranges from 5×10^4 to 5×10^6 (e.g., Yoder & Peale 1981; Peale 1999 and references therein). If we use 10^4 as a lower limit for Q , the eccentricity damping timescale for planet b would be $t_e \sim 4 \times 10^7$ yr. The tidal dissipation would not be able to prevent the eccentricity enhancement due to the secular perturbation. Nevertheless, we use the condition for survival against planet b's tidal inflation and disruption as a criterion to place constraints on the asymptotic values of e_b and the model parameters.

4.2. Evolution of the Eccentricities Due to Disk Depletion: Set A

We calculated the orbits of three planets during the depletion of the disk. The disk's contribution to the gravity decreases in accordance with equation (6). The largest influence of the disk depletion appears in the eccentricities of planets c and d. The changes of the eccentricities of three planets are shown in Figure 4. The solid, dotted, and dash-dotted lines show the evolution of planet b, c, and d, respectively.

These results indicate that the precession rates of periastra of the planets b, c, and d before the disk depletion are $120''$, $172''$, and $172''$ yr^{-1} respectively. In comparison with the normal mode formulation in Figure 2b, the precession rates of planets c and d, obtained through the numerical integration of their orbits, are nearly the same, whereas that of planet b is somewhat larger. This difference arises because in the analytic estimation, we consider a model in which planet b's precession circulates with respect to the other planets. But the numerical simulation shows that the periastra of the planets c and d are librating very close to each other and that of planet b is librating about both planet c and d with an amplitude $\varpi_b - \varpi_{c,d} \sim \pi$ prior to the disk depletion.

During the disk depletion, the evolution of the two outer planets (c and d) in the three-planet model is almost identical to what we found without taking into account the contribution of planet b. This result indicates that these planets are not signif-

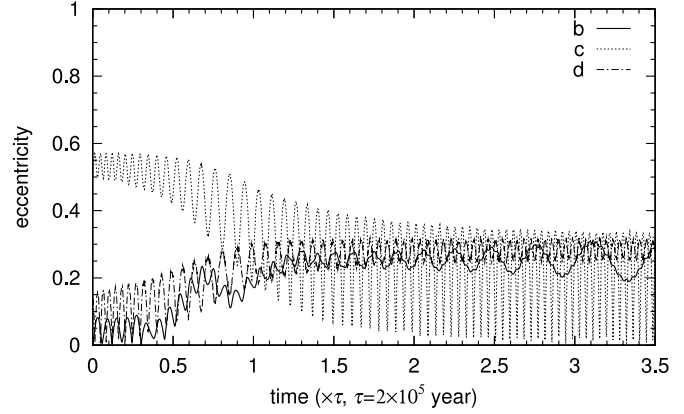


FIG. 4.—Evolution of the eccentricities of three planets around v And (set A) during the disk depletion. The solid, dotted, and dash-dotted lines show the evolution of planets b, c, and d, respectively.

icantly perturbed by planet b because it has a very limited supply of angular momentum. (To first order in e , secular perturbation can only lead to angular momentum transport.) Significant net exchange of the angular momentum between planets c and d starts at $t \sim 0.8\tau$ when they enter into a state of secular resonance with $A_{c,c} = A_{d,d}$. Through this net exchange of the angular momentum, e_d is enhanced while e_c is reduced. After the disk depletion, e_c and e_d attain their current observed values. Our analytical estimation of Figure 2a reproduces this net change well.

The average eccentricity of planet b increases as it interacts with the outer two planets. The oscillation period of planet b lengthens after $t \sim 2.5\tau$ to its current values as it is primarily influenced by planet d. In this model, the amplitude of e_b oscillates over a range of 0.2. At its present location and with such high asymptotic eccentricity, planet b would inflate and disrupt under the action of tidal dissipation.

4.3. Evolution of Planet b's Eccentricities Including the Post-Newtonian Gravity: Set B

When we add the post-Newtonian correction, the eigenfrequencies' present-day asymptotic value of g_3 increases by $67''$ yr^{-1} (eq. [8]). In Figure 1b, we showed that this effect is responsible for detuning the precession frequency of planet b such that it avoids being in a secular resonance with planets c and d today. However, during the early phase of their orbital evolution, the presence of the disk increases the magnitude of g_1 for the outer planets. Consequently, g_3 attains a value closer to g_1 at 1.3τ (see Fig. 3), and planet b enters into a secular resonance with planets c and d.

Figure 5 shows the evolution of all three planet's eccentricities obtained through the numerical integration of their orbit throughout the epoch of disk depletion. In these calculations, the effect of post-Newtonian modification on the host star's gravity is included. These results confirm that the eccentricity of planet b is excited at 1.3τ and eventually to an asymptotic value ~ 0.4 . With this large eccentricity, planet b is likely to be tidally disrupted during the subsequent circularization of its orbit.

4.4. Evolution of the Eccentricities Including the Stellar Rotation: Set C

With a contribution of stellar rotation to the overall potential, planet b precesses more rapidly. With a sufficiently large spin

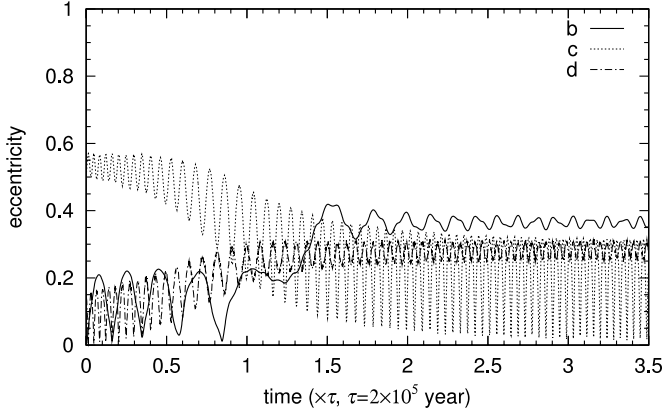


FIG. 5.—Same as Fig. 4, but for when the planet b is subjected to the post-Newtonian acceleration (set B).

frequency, this effect can shift the precession frequency of planet b well outside the secular resonance that was discussed above. In Figure 6, we plot the evolution of e_b for the host star's spin period of $S_* = 7, 3, 1$, and 0.7 days.

For the model with $S_* = 7$ days, the star's spin does not induce flattening sufficient to significantly modify the precession frequency of planet b. Our numerical results show that the evolution of the planetary system is essentially the same as Figure 5, i.e., e_b attains a large asymptotic value. In the model with $S_* = 3$ days, the amplitude of e_b grows to very large values before the disk is depleted. During the disk depletion, planet b becomes decoupled from this secular resonance at around 0.5τ , and the range of its eccentricity modulation becomes limited. The mean asymptotic eccentricity is ~ 0.6 . Although the magnitude of e_b depends sensitively on the timing when planet b

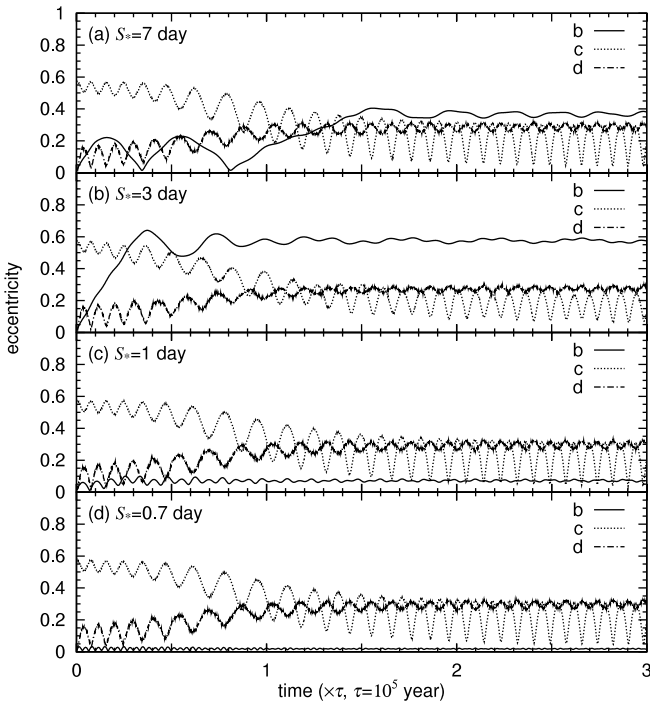


FIG. 6.—Evolution of the eccentricities of planets around v And (set C). The effects of the post-Newtonian gravity and flattened host star are included in the motion of the planet b. The meaning of the lines is the same as in Fig. 4. The rotation rate of the host star (S_*) is (a) 7.2, (b) 2.9, (c) 1, and (d) 0.7 days.

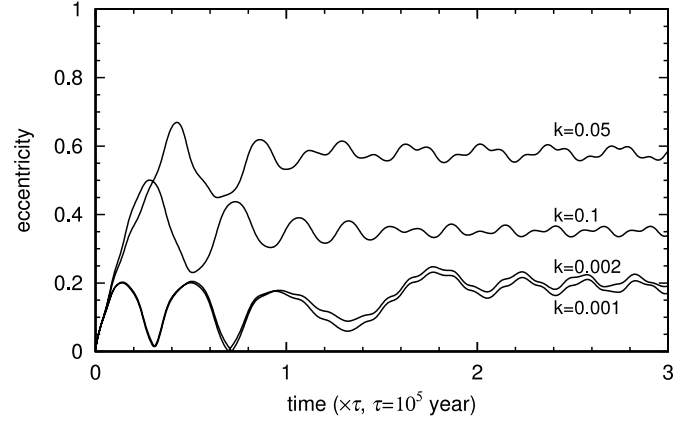


FIG. 7.—Evolution of eccentricity of planet b with different k_* .

becomes decoupled from the secular resonance, the asymptotic value generally exceeds 0.2 for a variety of prescriptions and boundary conditions. If the disk gas prevails over an extended period of time or the depletion is a slow process, planet b would be tidally disrupted well before the disk depletion is completed.

In the rapidly spinning-star model with $S_* = 1$ day, planet b's precession rate is markedly increased. Despite the passage of the first weak resonance in which $g_3 \simeq g_2$ at 0.2τ (Fig. 3), the excitation of e_b is limited to 0.1 (Fig. 6). During the remaining phase of disk depletion, planet b is no longer affected by any resonance such that e_b attains a relatively small asymptotic value. For the most rapidly spinning model with $S_* = 0.7$ days, the asymptotic value of e_b is even smaller. Based on these results, we find that $S_* < 2$ days is required in order for e_b to be always less than 0.2.

The apsidal advancement due to the stellar rotational distortion is $\propto k_* R_*^5 / S_*^2$ (see eqs. [9] and [12]). The sensitive dependence in the stellar radius introduces considerable uncertainties in the necessary condition for the survival of planet b. We consider the effect of pre-main-sequence contraction of v And in § 4.5. Physically, the magnitude of k_* is proportional to the Q -value, and its precise value is poorly known. Nevertheless, the results in Figures 6 and 7 can be scaled for different values of k_* , R_* , and S_* through the relation

$$\left(\frac{k_*}{0.02} \right) \left(\frac{R_*}{1.26 R_\odot} \right)^5 \left[\frac{S_*}{(0.7, 1, 3, 7) \text{ day}} \right]^{-2} = \text{const.} \quad (15)$$

For example, we show the evolution of e_b with different values of k_* in Figure 7. In this set of models, $R_* = 1.26 R_\odot$ and $S_* = 7$ days. For very small k_* (0.002 and 0.004), the effect of relativity dominates the contribution from the stellar rotation, and the results are similar to those presented in § 4.3. The model with $k_* = 0.1$ is equivalent to that with $S_* = 3$ days and $k_* = 0.02$. The evolution of e_b is almost the same as the case of $S_* = 3$ days in Figure 6. The result of a model with $k_* = 0.2$ is identical to the model with $S_* = 2$ days and $k_* = 0.02$. We found that negligible rotational distortion generally results in the excitation of the eccentricities for planet b.

4.5. Effect of the Stellar Contraction

Although the evolution of g_3 in Figure 3 can be applied to stars with different S_* , the relation in equation (17) must be satisfied on each line. During the pre-main-sequence evolution, the host star v And contracts with nearly constant angular momentum, $J = R_*^2 S_*$. Consequently, equation (17) is no longer

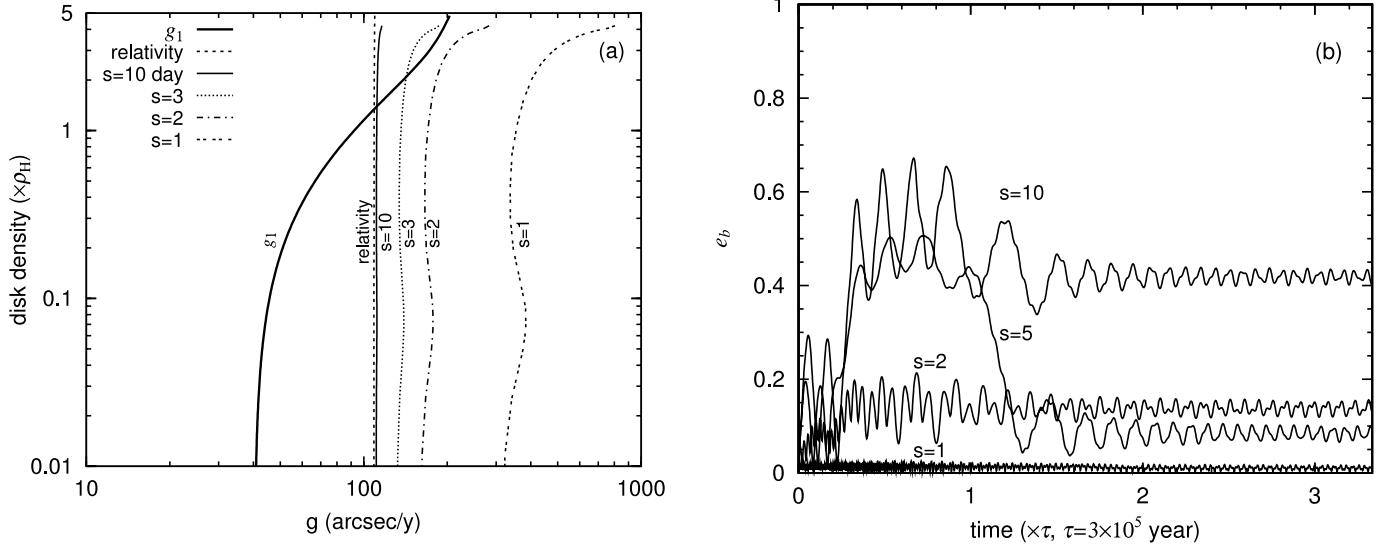


FIG. 8.—Evolution of planetary orbits during the disk depletion including the stellar contraction. Panel (a) shows changes of the eigenfrequencies of the system. The thick solid lines are g_1 . The dashed line is the case with the post-Newtonian effect. The solid, dotted, dash-dotted, and double-dashed lines are the cases of $S_* = 10.2, 3, 2$, and 1 day, respectively. In this plot, we use a real disk depletion time $\tau = 3 \times 10^6$ yr. Panel (b) illustrates the evolution of planet b's eccentricities with stellar rotation periods of $S_* = 10.2, 5, 2$, and 1 day.

satisfied, and $d\varpi_{i,\text{spin}}/dt \propto R_* J^2$ (see eq. [12]), which decreases during the stellar contraction. Increases in S_* correspond to a spin down and weaken the stabilizing contribution of stellar quadrupole moment so that the asymptotic values of e_b may be larger than that determined in Figure 3. Nevertheless, the reduction factor is limited by the linear dependence of $d\varpi_{i,\text{spin}}/dt$ on R_* .

The radius of the $1.3 M_\odot$ star ν And shrinks from 4 to $1.3 R_\odot$ during the first 30 Myr of its pre-main-sequence evolution. (We are indebted to Peter Bodenheimer for providing an evolution track for ν And.) This evolution timescale is comparable to the disk depletion timescale τ . However, the most rapid contraction phase occurs probably during the active phase of disk evolution, prior to the formation of planets b, c, and d. In order for all three planets, especially planet b, to acquire their large observed mass, the disk depletion presumably occurred after their formation. Thus, the range of S_* variation is unlikely to be large during the course of intense secular resonant interaction.

We also note that many classical and weak-line T Tauri stars have rapid rotation and relatively small S_* (Stassun et al. 1999). We assume the assumption that these diskless stars have already attained their main-sequence radius with an asymptotic spin period S_{*MS} . The results in Figure 3 indicate that the magnitude of g_1 decreases from $\sim 180''$ to $\sim 130'' \text{ yr}^{-1}$ within 1τ . Provided $S_{*MS} < 2$ days, $g_3 \gtrsim 150'' \text{ yr}^{-1}$, such that $g_3 > g_1$ throughout the evolution and the resonant condition may be avoided.

In order to illustrate the above conjecture, we carried out simulations with a prescription for the stellar contraction. Figure 8a shows g_b with different asymptotic values of S_{*MS} attained by ν And when R_* attains its asymptotic main-sequence value at $1.26 R_\odot$. We assume a disk depletion timescale of $\tau = 3 \times 10^6$ yr. Similar to the results in Figure 3, we derive the eigenfrequencies with equation (14). But for illustration simplicity, we neglected the contributions from terms that are second or higher order in e for these calculations. With this approximation, g_3 changes only due to the evolution of the stellar radius. From this figure, we found that the requirement for avoiding the secular resonance is $S_* < 2$ days at $1.26 R_\odot$. This analysis is verified by a series of

numerical simulations. With a set of direct orbital integrations, we show in Figure 8b the evolution of e_b for a range of S_* values. These results show that the asymptotic values of e_b are not significantly affected by the passage of the secular resonance during the disk depletion and that the planet can survive, provided S_{*MS} is shorter than about 1.5–2 days.

After the host star has already reached the main sequence, the spin period of ν And eventually increases from S_{*MS} to its present-day value (10.2 days) as angular momentum is lost through stellar wind via its open field lines. However, this process occurs on a timescale that is much longer than the stellar contraction and disk depletion timescales. When this process occurs, the inequality $g_3 > g_1$ is already well established, and the protracted stellar spin-down can no longer lead to secular resonance in this system.

4.6. Different Sets of Initial Conditions

In the above models, we adopt a set of initial conditions to reproduce the present kinematic structure of the planetary system around ν And. While the main purpose of this exercise is to demonstrate several potential physical effects that may affect the survivability of multiple planetary systems during the disk depletion, the exact course of evolution may depend sensitively on the initial conditions. In order to explore the robust aspects of the secular resonance passage, we consider here some variations in the initial conditions of the system.

4.6.1. Eccentricity Evolution of the ν And System Starting with Their Current Values

First, we adopt the simplest set of initial condition by imposing the planets, with their present-day orbital properties, in a disk environment. Figure 9 shows the evolution of the eccentricity of planets during the disk depletion in this hypothetical case. The top panel of this figure shows the evolution of e_b . The dash-dotted line is equivalent to the case in set A (i.e., with neither post-Newtonian force nor stellar rotation). The dashed line corresponds to the case in set B (with post-Newtonian force included). Four analogous cases of set C are shown by the

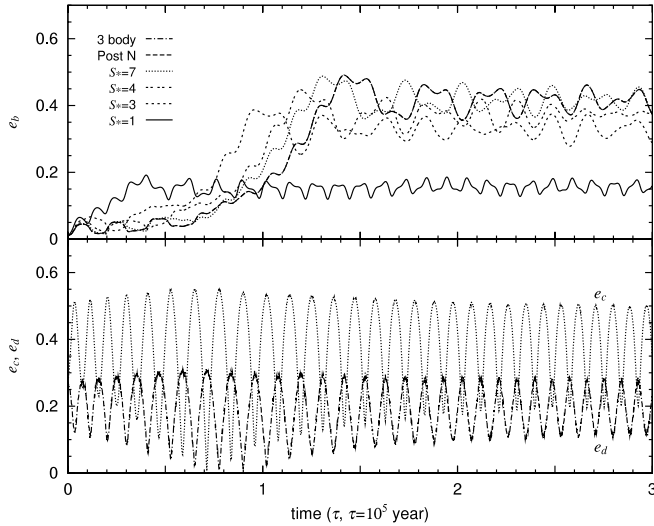


FIG. 9.—Evolution of the eccentricities starting with current orbital values. The top panel shows the eccentricity of planet b during the disk depletion. The dash-dotted line represents the case of set A, and the dashed line is for the case of set B. The dotted line, double-dashed line, short-dashed line, and solid line represent the cases of set C with $S_* = 7.2, 4.3, 2.9$, and 1 day, respectively. The bottom panel shows the eccentricities of the outer two planets c and d.

dotted line ($S_* = 7.2$ days), double-dashed line ($S_* = 4.3$ days), short-dashed line ($S_* = 2.9$ days), and solid line ($S_* = 1$ day). We did not include the effect of stellar contraction in these models.

The bottom panel shows evolution of e_c and e_d for the case in which $S_* = 2.9$ days. In comparison with the cases in §§ 4.2–4.4, the initial value of e_c in this new model is much smaller than that in the previous models. The final mean eccentricities are $e_c \sim 0.4$ and $e_d \sim 0.2$ in all cases. These sets of initial conditions do not lead to the present-day eccentricities of the outer two planets after the disk is depleted. They are merely chosen to illustrate some robust aspects of the conclusions we have reached above. In this set of models, the contribution of the disk to the overall potential modifies the precession frequencies of all planets such that the orbits of planets c and d no longer librate relative to each other. The misalignment of the periastra of the outer planets generally destabilizes the system on a long time scale.

Within the timescale of computation, the overall evolutionary tendency of e_b in this case is similar to those presented in §§ 4.2–4.4, that is, the asymptotic magnitude of e_b is much larger than the critical value (0.2) for tidal disruption. In these cases, the magnitude of e_b is excited when planets b and d pass through an phase of secular resonance. In the case of $S_* = 1$ day, the enhanced precession induces planet b to be initially in resonance with planet d. Because the outer planets' precession circulates with respect to each other and planet d's initial eccentricity is relatively large, the secular resonance g_2 is stronger than the analogous case in set C. The survival of planet b against this (g_2) resonant interaction in this case requires a slightly faster stellar rotation than the above standard cases in which the precessions of the outer planets librate with each other.

4.6.2. Migration of Planet d

In the models presented in §§ 4.1–4.6.1, we neglected tidal interaction between the planets and their nascent disks even though we assume that the disk structure is truncated by the

planets' tidal torque. In § 4.1, we have already provided justification of this assumption for planets b and c. Due to its proximity to the disk edge, the feedback of a perturbed disk can modulate planet d's eccentricity and semimajor axis through the dissipation of waves excited at its corotation and Lindblad resonances (Goldreich & Tremaine 1980; Goldreich & Sari 2003; Ogilvie & Lubow 2003). However, the finite residual amplitudes of e_c and e_d indicate that they were probably not severely damped. Depending on the inclination angle i , the observationally determined mass of planet d, $M_d = 3.75 M_J / \sin i$, may be sufficiently large for its resonant interaction with the disk to excite e_d (Artymowicz 1992). However, the asymptotic magnitude of e_d may be limited by nonlinear dissipation if the amplitude of planet d's epicycle becomes so large that it undergoes episodic excursion into the disk region outside the gap (Papaloizou et al. 2001). If such an equilibrium can be established, it may be sufficient to neglect this effect in the analysis of eccentricity evolution due to the secular resonance.

Planet-disk interaction with the tidally truncated outer regions of the disk would also lead to angular momentum loss from planet d through type 2 migration (Lin & Papaloizou 1986). The extent and rate of type 2 migration is determined by the poorly estimated angular momentum transfer rate in the disk. Extensive migration can lead to the formation of short-period planets such as 51 Peg (Lin et al. 1996) and planet b. But the present-day semimajor axis distribution of planets c and d suggests that they did not migrate over extensive regions in the disk (see § 4.1). This configuration requires the migration timescale to be shorter than the time interval between the formation of successive planets (in contrast to the resonant planets around 55 Cnc) and longer than or comparable to that associated with disk depletion (Ida & Lin 2004a).

In order to consider a range of limited migration, we introduce a prescription to examine the impact of type 2 migration on the secular interaction between the planets around ν And. For computational convenience, we adopt an ad hoc linear inward-migration prescription. Figure 10 shows the results of numerical orbital integration for a model in which planet d's semimajor axis a_d is set to decrease by 0.1 AU (Fig. 10a) and 1 AU (Fig. 10b). Since our objective is to highlight the effect of migration, the

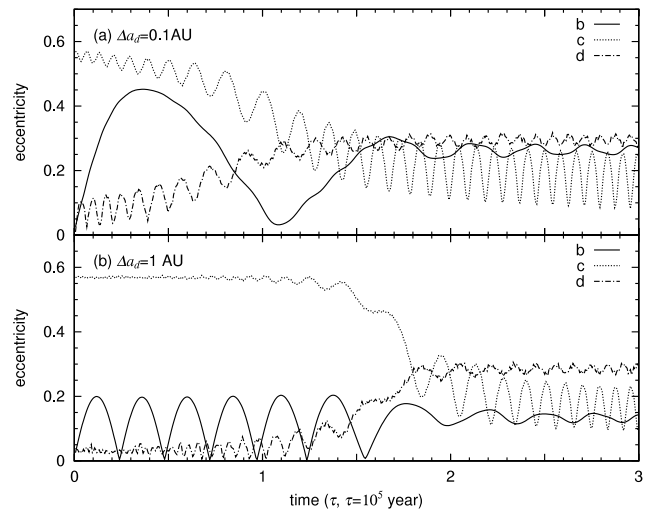


FIG. 10.—Evolution of ν And system in the case with planet d's migration. Panels (a) and (b) show the evolution of eccentricities in the cases that migration of the planet d is 0.1 and 1 AU, respectively. The meaning of the lines is the same as in Fig. 4.

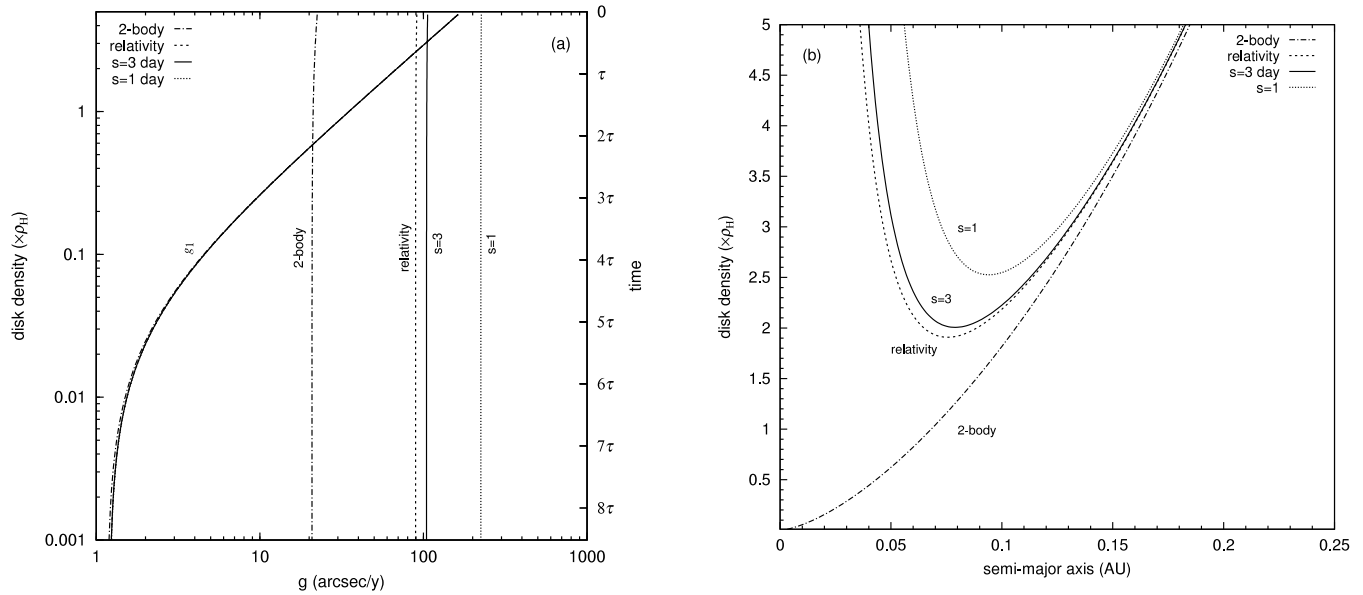


FIG. 11.—Evolution of a two-planet system during disk depletion. Panel (a) shows the changes of the eigenfrequencies of the two-planet system during the disk depletion. The thick solid lines, g_1 , are almost identical to the precession rate of the outer planets. The vertical lines are the frequencies associated with the inner short-period planet. The thick dash-dotted line is for the two-planet interaction. The dashed line is the case with the post-Newtonian effect. The solid and dotted lines are the case of $S_* = 3$ and 1 days, respectively. Panel (b) illustrates the location of secular resonance caused by the outer planet vs. disk density.

stellar spin rate is assumed to be that of the present-day value at all times. We also do not modify the location of the disk edge from 4.5 AU in these models. Our results do not sensitively depend on the details of these prescriptions.

With these initial conditions we integrate the system with the decay of a_d over 3τ . Our models indicate that with a wider initial separation, the weakened secular perturbation between planets c and d induces lower amplitude eccentricity modulation and a smaller precession rate during the early evolutionary phases. However, with a larger initial semimajor axis, planet d comes closer to the disk edge and therefore attains a larger disk potential-induced precession rate.

During planet d's inward migration, the fractional contribution to $\dot{\varpi}$ due to the planets' mutual secular perturbation grows with respect to the precession due to the disk potential. Similar increases also occur during the disk depletion. Our results reproduce the present-day kinematic properties of planets c and d with diverse asymptotic values of e_b . Although g_3 does not depend explicitly on a_d , its evolution is affected by g_1 , especially when the system enters into a resonance. In the migrating-planet models, the initial value of g_1 is reduced such that the passage of the secular resonance (with $g_1 \simeq g_3$) occurs at a slightly later epoch and that the amplitude of e_b excited by the secular resonance in this slowly spinning ($S_* = 10.2$ days) model (Fig. 10a) becomes larger than that with $S_* = 7$ days in set C. This effect is more pronounced in the model with a large migration extent (see Fig. 10b). The reduced value of g_1 improves the survivability of planet b around its host star with a longer spin period.

From an analysis similar to that in § 3.5, we found that for the migrating planet models, secular resonance would be permanently avoided with $g_3 > 150'' \text{ yr}^{-1}$. This condition is achieved with $S_* < 3$ days. But we note that the stability of ν And depends sensitively on the current libration properties of planets c and d. Extensive migration of planet d after the disk dissipation may lead to dynamical instabilities. The basic conclusion in this section is that even with the migration of planet d, the survival

of planet b requires the initial spin period of ν And to be much shorter than its present-day value.

4.7. Model Case of Two-Planet System

In all the models we have presented above, we focused on the kinematic structure of the three-planet system around ν And. Our analysis indicates that although three eigenfrequencies exist in this complex system, the survival of planet b only depends on the resonant condition $g_1 \simeq g_3$. The passage of the weak resonance $g_2 \simeq g_3$ did not lead to major changes in e_b . We now demonstrate that one outer planet may be adequate to induce the passage of secular resonance, which may lead to the tidal disruption of short-period planets. Since (1) the formation of the first gas giant planet can promote the formation of additional planets, (2) most extrasolar planets are multiple Jovian-mass planets, and (3) many short-period planets have long-period siblings, the demonstration of general vulnerability of short-period planets in systems with two or more planets provides supporting evidence that a large fraction of planets that migrated inward perished during the epoch of disk depletion.

We now consider a system with one modest-period ($P \sim 1$ yr) and one short-period planet ($P \lesssim 1$ month). We show that the condition for secular resonance generally occurs during the disk depletion and that if the outer planet has a larger mass and a modest or large eccentricity, the passage of the secular resonances would excite the eccentricity of the short-period planet. In this case, tidal dissipation within the short-period planet would lead it to inflation and disruption.

If we neglect the effects of post-Newtonian potential and stellar rotation, the secular resonance due to the outer planet would sweep across the entire disk, and all the short-period planets would be affected by the passage of the secular resonance. The survival condition for the short-period planets is that their precession rate $\dot{\varpi}$ must be faster than that of the outer planet prior to the disk depletion so that the sweeping secular resonance may be affected. In Figure 11a, we show frequencies g of a model case of a two-planet system. Here, we arbitrarily choose the values of

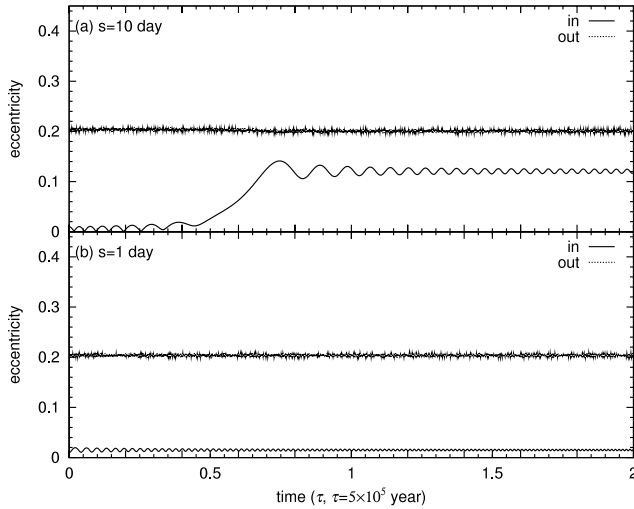


FIG. 12.—Example of the eccentricity evolution of the two-planet system. The inner planet is located at 0.05 AU, and the outer planet is at 1 AU. Panel (a) shows the case with a 10 day stellar rotation period. Panel (b) illustrates the case with a 1 day stellar rotation.

the inner planet to be $M_1 = 0.5M_J$ and $a_1 = 0.05$ AU and those of the outer planet to be $M_2 = 2M_J$ and $a_2 = 1$ AU. The host star is $1 M_\odot$ and $1 R_\odot$. We assumed the outer disk edge at 3 AU. In this hypothetical system, $S_* \lesssim 1$ day is required.

We plot the location of the resonance due to the outer planet in Figure 11b. Without the post-Newtonian effect and stellar rotation (*dash-dotted line*), the resonance sweeps across the entire disk to the surface of the host star. In reality, the post-Newtonian potential modifies the precession frequency of the short-period planet, and those planets with a semimajor axis less than 0.03 AU can avoid the resonance. Outside 0.05 AU, the post-Newtonian effect is relatively weak. Nevertheless, a rapidly rotating star may introduce sufficient distortion to its mass and gravity distribution that the precession of the short-period planet may avoid secular resonance.

Figure 12 shows results of numerical simulation in this system. We use initial eccentricities of $e_1 = 0.01$ and $e_2 = 0.2$. In the case of $S_* = 1$ day, the small eccentricity of planet b can be preserved during the passage of the secular resonance. But it is excited to $e_b > 0.1$ in the case of $S_* = 10$ days. Based on these results, we postulate that stars with rapid initial spins may retain more short-period planets in multiple systems than slow rotators.

5. SUMMARY AND DISCUSSIONS

We investigate the orbital evolution of the planets around ν And during the depletion of the protostellar disk, taking into account the post-Newtonian gravity and the rotational distortion of the host star.

The semimajor axis of planet b around ν And is ~ 0.06 AU. This location is close to the secular resonance with planet c and d. Around a point-mass potential and under the action of secular perturbation from the outer planets, the eccentricity of planet b would be excited to an amplitude ≥ 0.2 . With such a large eccentricity, the tidal dissipation of energy during its orbital circularization would induce inflation of the short-period planet on the disk depletion timescale of $\sim 10^6$ yr. Tidal inflation continues until the short-period planet fills its Roche lobe, loses mass, and migrates away from its host star (Gu et al. 2003).

Despite this disruptive tendency, ν And b survives today with its present low eccentricity due to the advancement of its periastron, which arises from the post-Newtonian correction on the gravity of its host star. However, this offset in the precession frequency of ν And b alone is not adequate to prevent the passage of the sweeping secular resonance and the substantial growth of its eccentricity during the depletion of the disk.

During the epoch of disk depletion, most host stars are rapid rotators. The rotational flattening of the host star introduces a J_2 component in the moment of the stellar gravitational potential. The precession caused by this effect is also important in determining the evolutionary fate of the short-period planets.

Using a numerical integrator, we calculate the orbital evolution of planet b during the epoch of disk depletion. For computational simplicity, we adopt a boundary condition in which the inner region of disk, where the planets are orbiting today, is assumed to be completely cleared. But beyond the orbit of ν And d, we adopt the initial surface density of the disk to be 5 times that of minimum mass solar nebula model. Since the process of the disk depletion changes the eccentricities of the outer two planets, we take the initial eccentricities of the outer two planets to be $e_c \sim 0.57$ and $e_d \sim 0.04$. These parameters lead to the current eccentricities of these two outer planets.

From the numerical integration, we found that the planet b is near the secular resonance prior to the disk depletion as long as the rotation period of its host star is longer than 2 days. For example, if ν And always had its present rotation period of 10.2 days, the passage of the secular resonance during the early phase of disk depletion would have excited the eccentricities of planet b to $e_b \gtrsim 0.4$. It is possible for planet b to avoid secular resonance altogether if its initial spin period is less than 2 days.

We used the disk depletion time of order of 10^5 yr in the numerical simulations that we presented here. These results indicate that the duration of enhanced eccentricity amplitude is relatively brief. In the actual disks, the depletion timescale is 10^6 – 10^7 yr, and planet b attains a large eccentricity for a much longer duration. In a recent simulation for the formation and migration of protoplanets, Ida & Lin (2004a) showed that the expected fraction of planets with periods of less than a week to between 1 and 2 months is much higher than that actually observed. They suggested that most short-period planets may have perished during their infancy. Since most host stars with known extrasolar planets also have signs of additional planets, and since a large fraction of young stellar objects have spin periods longer than a few days, we suggest that short-period planets may be highly vulnerable to this type of disruption process. Note that during this early epoch, solar-type stars remain mostly convective so that the accretion of a tidally disrupted protoplanetary material would not significantly alter their surface metallicity.

Our main task is to account for the survival of ν And b. If the initial spin period of ν And is comparable to or less than 1 day, planet b would be able to avoid the secular resonance of the outer planets during the disk depletion. The apsidal precession rate of planet b is less than that of c and d before the disk depletion, provided $S_* > 2$ days. During the disk depletion, the precession rate of the periastron of planet b is slightly decreased. This is mainly because of the eccentricity change of planet c. Since planet b is far from the inner edge of the tidally truncated disk, its depletion does not change planet b's precession rate. However, the mean advance rate of the librating

periastra of the outer planets decreases greatly from $200'' \text{ yr}^{-1}$ to $50'' \text{ yr}^{-1}$. That is, the relation of $d\varpi_b/dt > d\varpi_{c,d}/dt$ is maintained, provided $S_* < 2$ days. This condition also shows that there would be no problem even if the spin-down occurs after the outer two planets have exchanged the angular momentum. This result is not sensitive to early disk density. Between $10\rho_H$ and $2\rho_H$ in Figure 3, the relation of g_3 and g_1 is insensitive to the disk mass. Since the resonance of planets c and d occurs at around the density of the minimum mass disk ($\propto \sin I$), more massive disks lead the secular resonance unless the host star rotates rapidly. Based on these results, we infer from the pres-

ence of planet b that the spin period of ν And was shorter than 3 days during the early epochs of its evolution.

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APPENDIX

SECULAR RESONANCE AND EIGENFREQUENCIES OF THE SYSTEM

The time-averaged precession rate is close to one of the eigenfrequencies of the system in many cases, especially when the planets' orbits circulate with respect to each other. However, the precession rates alone do not provide adequate information on the condition of the secular resonance. Approximations based on the lowest order expansion in eccentricity are also not sufficient for the analysis of a system like ν And in which the eccentricity of planet c is quite high (~ 0.7) prior to the disk depletion. In this section, we estimate the eigenfrequencies of the three-planet system and consider their high-order apsidal motion in order to interpret the results of our numerical simulation in § 4.

The disturbing functions related to secular perturbation of two planets (1 and 2 with semimajor axis $a_1 \ll a_2$) are written as (e.g., Brouwer & Clemence 1961; Murray & Dermott 1999)

$$R_1 = \frac{GM_2}{a_2} \frac{a_2}{r_{12}}, \quad (\text{A1})$$

$$R_2 = \frac{GM_1}{a_2} \frac{a_2}{r_{12}}, \quad (\text{A2})$$

where r_{12} is the distance between planets 1 and 2. The mutual distance can be expanded in terms of the eccentricity and the semimajor axis. For illustrative purpose, we only consider a two-dimensional system here. In this case, to second order in the eccentricity, a secular variation in the mutual distance between the two planets is

$$\begin{aligned} \frac{a_2}{r_{12}} = & \frac{1}{2} b_{1/2}^{(0)}(\alpha) + \frac{1}{2} (e_1^2 + e_2^2) F_1(\alpha) + \frac{1}{2} e_1^2 e_2^2 F_2(\alpha) + \frac{1}{4} e_1^4 F_7(\alpha) + \frac{1}{4} e_2^4 F_8(\alpha) \\ & + [F_3(\alpha) + e_1^2 F_4(\alpha) + e_2^2 F_5(\alpha)] e_1 e_2 \cos(\varpi_1 - \varpi_2) + \frac{1}{2} e_1^2 e_2^2 F_6(\alpha) \cos(2\varpi_1 - 2\varpi_2), \end{aligned} \quad (\text{A3})$$

where $\alpha = a_1/a_2 < 1$. The functions F_1 to F_8 are expressed in terms of the complete integral of the first kind, $F(\alpha)$, and the second kind, $E(\alpha)$.

$$F_1 = \frac{1}{\pi(1-\alpha^2)^2} [(\alpha^2 - 1)F + (\alpha^2 + 1)E], \quad (\text{A4})$$

$$F_2 = \frac{3\alpha^2}{4\pi(1-\alpha^2)^4} [(\alpha^2 - 1)(3\alpha^2 + 5)F + 8(\alpha^2 + 1)E], \quad (\text{A5})$$

$$F_3 = \frac{-1}{\pi\alpha(1-\alpha^2)^2} [(1-\alpha^2)(\alpha^2 - 2)F + 2(\alpha^4 - \alpha^2 + 1)E], \quad (\text{A6})$$

$$F_4 = \frac{-1}{8\pi\alpha(1-\alpha^2)^4} [(\alpha^2 - 1)(33\alpha^4 - 13\alpha^2 + 4)F + (9\alpha^6 + 50\alpha^4 - 15\alpha^2 + 4)E], \quad (\text{A7})$$

$$F_5 = \frac{-\alpha}{8\pi(1-\alpha^2)^4} [(1-\alpha^2)(\alpha^2 - 9)(2\alpha^2 + 1)F + (4\alpha^6 - 15\alpha^4 + 50\alpha^2 + 9)E], \quad (\text{A8})$$

$$F_6 = \frac{3}{8\pi\alpha(1-\alpha^2)^4} [(\alpha^2 - 1)(\alpha^4 + 9\alpha^2 - 2)F - 2(\alpha^2 + 1)(\alpha^4 - 6\alpha^2 + 1)E], \quad (\text{A9})$$

$$F_7 = \frac{1}{8\pi(1-\alpha^2)^4} [(\alpha^2 - 1)(27\alpha^4 - 5\alpha^2 + 2)F + 2(3\alpha^6 + 23\alpha^4 - 3\alpha^2 + 1)E], \quad (\text{A10})$$

$$F_8 = \frac{1}{8\pi(1-\alpha^2)^4} [(\alpha^2 - 1)(-\alpha^4 + 19\alpha^2 + 6)F + 2(\alpha^6 - 3\alpha^4 + 23\alpha^2 + 3)E]. \quad (\text{A11})$$

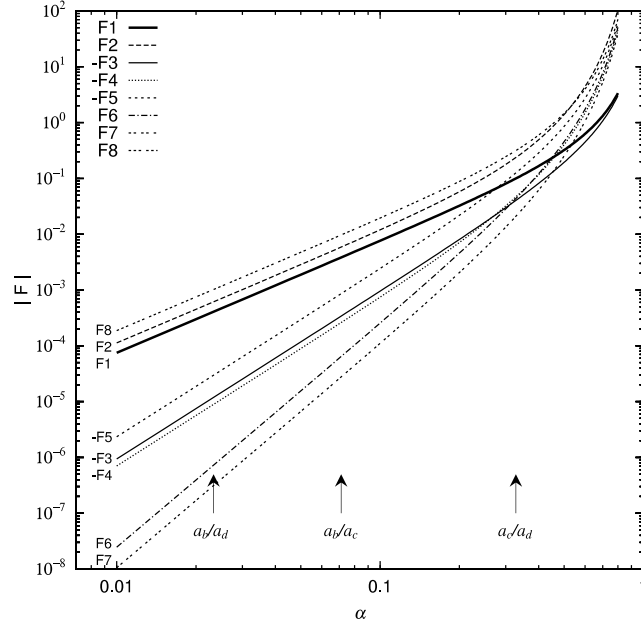


FIG. 13.—Values of $|F|$ vs. the ratio of the semimajor axes (α). Arrows indicate the values of v And planets.

Values of these functions are shown in Figure 13. These functions can also be written using the Laplace coefficient such that $F_1 = \alpha b_{3/2}^{(1)}/4$ and $F_3 = -\alpha b_{3/2}^{(2)}/4$. Adopting the valuables $H_i = (n_i M_i)^{1/2} a_i e_i \sin \varpi_i$ and $K_i = (n_i M_i)^{1/2} a_i e_i \cos \varpi_i$, the equations of motion are written as

$$\frac{dH_i}{dt} = \sqrt{1 - e_i^2} M_i \frac{\partial R}{\partial K_i}, \quad (\text{A12})$$

$$\frac{dK_i}{dt} = -\sqrt{1 - e_i^2} M_i \frac{\partial R}{\partial H_i}. \quad (\text{A13})$$

Substituting the disturbing function in these equations, we get

$$\frac{dH_i}{dt} = A_{i,i} K_i + A_{i,j} K_j, \quad (\text{A14})$$

$$\frac{dK_i}{dt} = -A_{i,i} H_i - A_{i,j} H_j, \quad (\text{A15})$$

where $A_{i,i}$ and $A_{i,j}$ are

$$A_{i,i} = \frac{GM_j}{n_i a_i^2 a_j} \left[\left(1 - \frac{1}{2} e_i^2 \right) F_1 + e_i^2 F_7 + e_j^2 (F_2 - F_6) + 2F_4 e_i e_j \cos(\varpi_i - \varpi_j) \right], \quad (\text{A16})$$

$$A_{i,j} = \frac{G}{a_i a_j^2} \sqrt{\frac{M_i M_j}{n_i n_j}} \left[\left(1 - \frac{1}{2} e_i^2 \right) F_3 + e_i^2 F_4 + e_j^2 F_5 + 2F_6 e_i e_j \cos(\varpi_i - \varpi_j) \right], \quad (\text{A17})$$

for the inner planet, and

$$A_{i,i} = \frac{GM_j}{n_i a_i^3} \left[\left(1 - \frac{1}{2} e_i^2 \right) F_1 + e_i^2 F_8 + e_j^2 (F_2 - F_6) + 2F_5 e_i e_j \cos(\varpi_i - \varpi_j) \right], \quad (\text{A18})$$

$$A_{i,j} = \frac{G}{a_i^2 a_j} \sqrt{\frac{M_i M_j}{n_i n_j}} \left[\left(1 - \frac{1}{2} e_i^2 \right) F_3 + e_i^2 F_5 + e_j^2 F_4 + 2F_6 e_i e_j \cos(\varpi_i - \varpi_j) \right], \quad (\text{A19})$$

for the outer planet. Although $A_{i,i}$ and $A_{i,j}$ include H_i and K_i as $e_i^2 = (H_i^2 + K_i^2)/(M_i n_i a_i^2)$, we treat the eccentricities in $A_{i,i}$ and $A_{i,j}$ as constant corrections and solve the equations. In addition, we use a time-averaged value for $\cos(\varpi_i - \varpi_j)$ in $A_{i,i}$ and $A_{i,j}$. We set

it to be zero when the planets' orbits are circulating relative to each other. The terms with $e^2 \cos$ in $A_{i,i}$ are not large compared with other terms, since $F_4 \ll F_1$ and F_2 . In the v And system, the terms with $e^2 \cos$ in $A_{i,j}$ are also not large because $F_6 < |F_3|$, $|F_4|$, and $|F_5|$. The outcome does not change significantly, regardless of whether the planets' orbits are circulating or librating.

From the numerical simulations of Nagasawa et al. (2003), we find that planets c and d maintain the coincidence of periastron of $\langle \cos(\varpi_c - \varpi_d) \rangle \sim 0.9$ during the disk depletion. The equations of motion (A15) for three interactive planets can be generalized such that

$$\frac{dH_i}{dt} = A_{i,i}K_i + A_{i,j}K_j + A_{i,\ell}K_\ell, \quad (\text{A20})$$

$$\frac{dK_i}{dt} = -A_{i,i}H_i + A_{i,j}H_j + A_{i,\ell}H_\ell. \quad (\text{A21})$$

The eigenfrequencies of the system, g , are obtained from a determinant,

$$\begin{vmatrix} A_{i,i} - g & A_{i,j} & A_{i,\ell} \\ A_{j,i} & A_{j,j} - g & A_{j,\ell} \\ A_{\ell,i} & A_{\ell,j} & A_{\ell,\ell} - g \end{vmatrix} = 0. \quad (\text{A22})$$

The secular resonance occurs when one of the eigenfrequencies becomes closer to another eigenfrequency. The solutions of equation (A21) have forms of

$$H_i = \sum_{j=1}^3 M_i^j \sin(g_j t + \beta_j), \quad (\text{A23})$$

$$K_i = \sum_{j=1}^3 M_i^j \cos(g_j t + \beta_j), \quad (\text{A24})$$

where β_j and two values of M_i^j are constants of the integration. Four values of M_i^j can be written by g_j and the other two M_i^j .

The precession rate of the periastron is given by

$$\frac{d\varpi_i}{dt} = \frac{(1 - e_i^2)^{1/2}}{n_i a_i^2 e_i} \frac{\partial R_i}{\partial e_i}. \quad (\text{A25})$$

The rate of precession caused by an outer planet j to an inner planet i is

$$\frac{d\varpi_i}{dt} = \frac{GM_j(1 - e_i^2)^{1/2}}{n_i a_i^2 a_j} \left[F_1 + e_j^2 F_2 + e_i^2 F_7 + \frac{e_j}{e_i} (F_3 + 3e_i^2 F_4 + e_j^2 F_5) \cos \eta + F_6 e_j^2 \cos(2\eta) \right], \quad (\text{A26})$$

and that caused by an inner planet j to an outer planet i is

$$\frac{d\varpi_i}{dt} = \frac{GM_j(1 - e_i^2)^{1/2}}{n_i a_i^3} \left[F_1 + e_j^2 F_2 + e_i^2 F_8 + \frac{e_j}{e_i} (F_3 + 3e_i^2 F_5 + e_j^2 F_4) \cos \eta + F_6 e_j^2 \cos(2\eta) \right], \quad (\text{A27})$$

where $\eta = \varpi_i - \varpi_j$. From Figure 13, we can see that F_2 and F_8 have larger values than F_1 and that F_5 is larger than F_3 . Therefore, in the system with large eccentricities, the analytic apsidal motion in equation (2) in the text deviates from the numerical simulations.

The averaged precession rate is almost equal to $A_{i,i}$, provided the planets are circulating ($\langle \cos \eta \rangle \sim 0$) and have low eccentricities. In addition, near the secular resonance $A_{i,i} \sim A_{j,j}$. This is the reason we sometimes associate secular resonance with the coincidence of two planets' apsidal motion. However, this is a description with some assumptions. In a librating orbit, two planets synchronize because the $\cos \eta$ term exceeds the other terms, while $A_{i,i}$ and $A_{j,j}$ are not necessarily close.

Contribution from the disk and other physical effects to the disturbing function can be added linearly to equation (A2) in our scheme. From equation (4), the secular term of the disturbing function of the disk is

$$R_{\text{disk},i} = 4\pi G \Sigma(r_i) a_i \sum_{n=0} \left\{ \frac{B_n e_i^2}{2} \left[1 + \frac{1}{8} (2n-1)(n-2) e_i^2 \right] \left(\frac{r_i}{d} \right)^{2n+1/2} \right\}. \quad (\text{A28})$$

Since this is an axisymmetric perturbation (that is, the equation does not contain ϖ), only $A_{i,i}$ is modified among $A_{i,i}$ and $A_{i,j}$. In this case, the contribution to the $A_{i,i}$ is the same as the contribution to the apsidal motion,

$$A_{\text{disk},i,i} = \frac{d\varpi_i}{dt} = 4\pi G \Sigma(r_i) a_i \sum_{n=0} \left\{ B_n \left[1 + \frac{1}{4} (n-1)(2n-1) e_i^2 \right] \left(\frac{r_i}{d} \right)^{2n+1/2} \right\}. \quad (\text{A29})$$

Similarly, the contribution of the stellar rotation is

$$A_{\text{spin},i,i} = \frac{d\varpi_i}{dt} = \frac{k_* \Omega_*^2}{n_i} \left(\frac{R_*}{a_i} \right)^5 \left(1 + \frac{m_i}{M_*} \right) (1 + 2e_i^2). \quad (\text{A30})$$

But the effects of the stellar rotation and the post-Newtonian gravity only affect the short-period planet. Since we consider the eccentricity of planet b as negligible in the main text, $A_{i,i}$ is modified as

$$A'_{i,i} = A_{i,i} + A_{\text{disk},i,i} + \frac{k_* \Omega_*^2 R_*^5 (1 + M_i/M_*)}{n_i a_i^5} + \frac{3n_i G M_*}{a_i c^2}. \quad (\text{A31})$$

In putting this $A'_{i,i}$ in equation (A22) instead of $A_{i,i}$, we can get three eigenfrequencies of the system.

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